



LASER SECURITY SYSTEM

BEAM-BREAK INSTRUCTION DETECTION USING ARDUINO UNO

TEAM VMTS

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Guide: Sandhya Pokharkar • 15 July 2026

PROJECT BACKGROUND & PROBLEM

WHY THIS PROJECT

- Affordable security for homes & small businesses
- CCTV & PIR sensors are costly or unreliable
- Precise, line-of-sight detection needed
- Applies embedded systems knowledge practically

Real-world basis: theft & unauthorized access at homes, shops, schools & warehouses – a recurring concern in unmonitored areas.



PROBLEM STATEMENT

Existing security systems are too expensive, complex, or unreliable for widespread use – leaving entry points vulnerable to undetected intrusions.

Current Gap: CCTV needs constant monitoring; PIR gives frequent false alarms.

Impact: Rising theft & reliance on costly manual guarding.

OBJECTIVES & EXISTING SYSTEMS



Real-time tripwire monitoring
at entry points



Low-latency alert on beam
interruption



Minimize false alarms vs. PIR
sensors



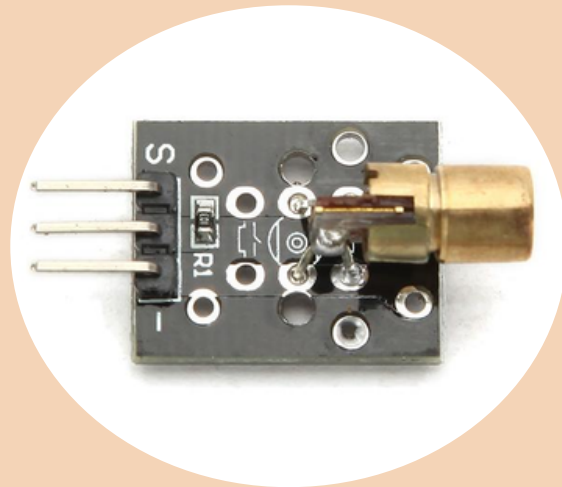
Low-cost, easy-to-install
solution

HOW TODAY'S ALTERNATIVES COMPARE

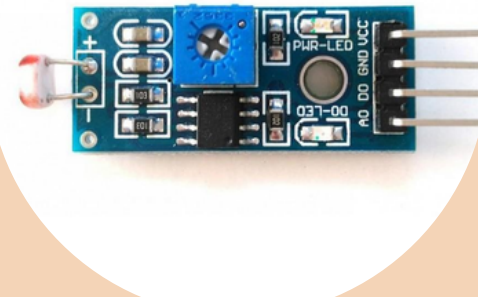
FEATURE	CCTV	PIR SENSORS	LASER (PROPOSED)
Cost	High	Medium	LOW
False Alarms	Low	High	VERY LOW
Power Use	High	Medium	LOW
Precision	Medium	Low	HIGH

PROPOSED SOLUTION & INNOVATION

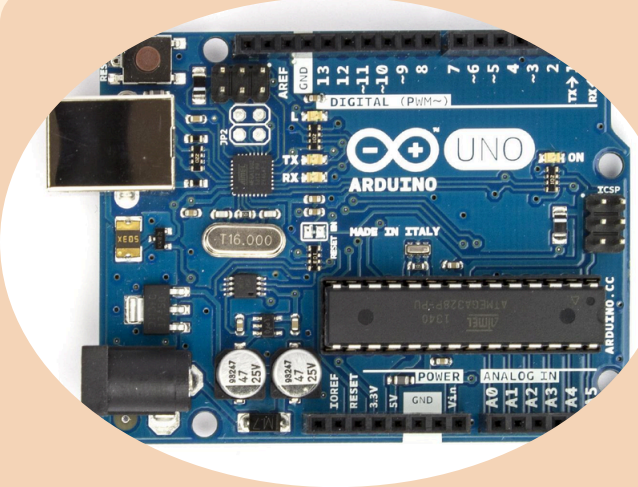
A laser diode emits a continuous beam across an entry point onto an LDR sensor. When the beam is interrupted, the LDR reading changes sharply and Arduino instantly triggers a buzzer alarm.



Laser Module
Emits beam



LDR Sensor
Detects break



Arduino Uno
Processes signal



Buzzer
Sounds alarm

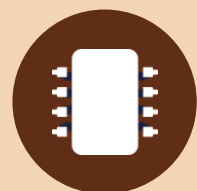
Innovation: turning a traditionally "dumb" tripwire into a smart, connected alert system – extensible toward IoT, cloud logging & app alerts.

HARDWARE ARCHITECTURE & CIRCUIT

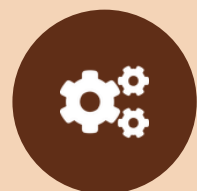
BLOCK FLOW



Laser Module – Transmits beam



LDR Sensor – Detects beam



Arduino Uno – Processes signal



Buzzer – Sounds alarm

CIRCUIT CONNECTIONS

Laser VCC / GND	5V / GND
LDR Leg 1	5V (10kΩ divider)
LDR Leg 2	Analog Pin A0
Buzzer +/-	Digital Pin 8 / GND
USB Cable	Powers Arduino

Power: 5V, ~100–200 mA total • Logic: 5V • LDR read: 0–5V analog

COMPONENTS & SOFTWARE ARCHITECTURE

Component	Qty	Purpose
Arduino Uno	1	Main controller
Laser Module	1	Emits beam
LDR	1	Detects interruption
Breadboard	1	Prototyping platform
Buzzer	1	Sounds alarm
Jumper Wires	As needed	Connections



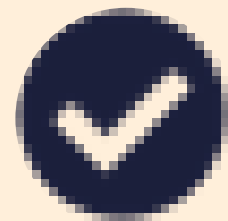
SOFTWARE STACK

Language: Arduino C/C++
Data Storage: JSON file
Repository: Not yet published

No external libraries, APIs, or cloud services at this prototype stage – runs on built-in Arduino functions.

WORKING PRINCIPLE & ALGORITHM

The laser projects a steady beam onto the LDR. While intact, the buzzer stays OFF. When blocked, the LDR reading drops sharply and Arduino instantly activates the buzzer.



BEAM INTACT

LDR reads high – buzzer OFF



BEAM BROKEN

LDR drops sharply below threshold



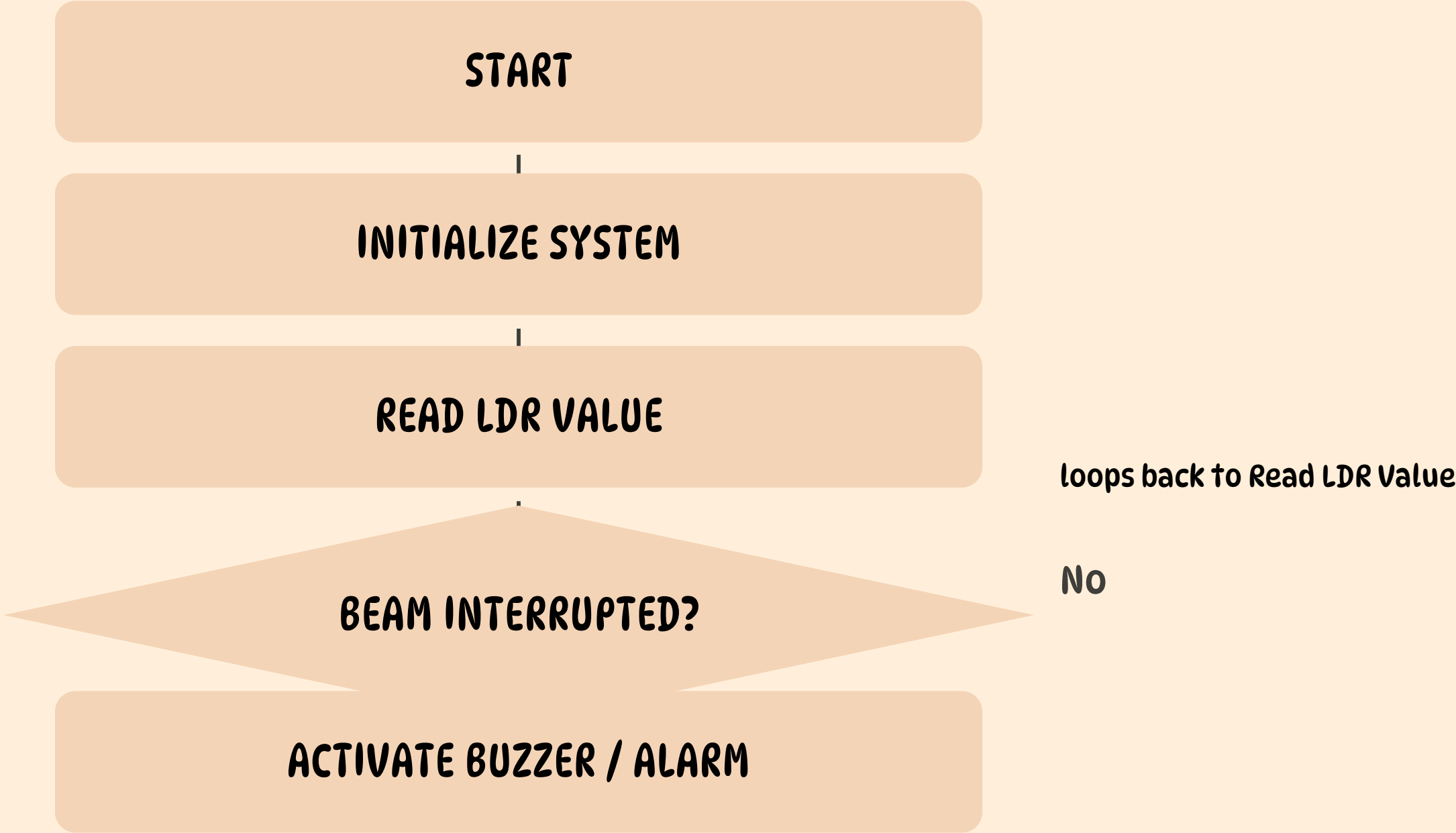
ALERT TRIGGERED

Arduino activates buzzer instantly

ALGORITHM

- Initialize pins (LDR→A0, Buzzer→D8)
- Continuously read LDR analog value
- Compare against calibrated threshold
- Below threshold → activate buzzer
- Above threshold → keep buzzer OFF
- Repeat loop continuously

SYSTEM FLOWCHART



Yes → Buzzer / Alarm activates. The system loops continuously, re-checking beam status after every read cycle.

TESTING & RESULTS

TESTING SUMMARY

Method: Manually interrupting the beam, observing buzzer
Conditions: Indoor lighting, varying distances
Result: Buzzer activated reliably every time
Failures Noted: Occasional false trigger in strong ambient light – resolved via calibration



NEAR-INSTANT

Response Time



HIGH

Accuracy



CONSISTENT

Reliability

ADVANTAGES & APPLICATIONS

ADVANTAGES



Reliable – consistent beam-break detection



Instant alerts, no manual monitoring



~₹805 total, cheaper than CCTV



High precision, minimal false alarms

APPLICATIONS



Homes & Shops



Industrial Warehouses



Educational Labs



Medical Storerooms

CONCLUSION & FUTURE SCOPE

The Laser Security System proves reliable, real-time intrusion detection doesn't require expensive infrastructure – a simple laser-LDR beam-break mechanism with Arduino Uno delivers fast, accurate, low-cost monitoring.



IoT Connectivity



AI Filtering



Mobile App Alerts



Cloud Logging

THANK YOU

Team VMTS | Vidyamandir Manda Titwala | Guide: Sandhya Pokharkar • Questions & Feedback Welcome